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Empowered or Replaced?

UX Designers' Perceptions of AI in the Workplace Through the Lens of Self-Efficacy

A Doctoral Dissertation

Presented to

The Dissertation Committee of the School of Economics and Business Administration

Saint Mary's College of California

In Partial Fulfillment

of the Requirements for the Degree

Doctor of Business Administration

By

Christine Lee

April 2025

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This doctoral dissertation, written under the direction of the candidate's dissertation committee and approved by members of the committee, has been presented to and accepted by the dissertation committee, in partial fulfillment of the requirements for the Executive Doctorate of Business Administration degree.

  
Christine Lee (Apr 23, 2025 17:15 PDT)

04/23/2025

Candidate: Christine Lee

Date

Dissertation Committee:

  
Michal Strahilevitz (Apr 23, 2025 17:16 PDT)

04/23/2025

Advisor: Dr. Michal Strahilevitz, Ph.D.

Date

  
michae H (Apr 23, 2025 21:32 EDT)

04/23/2025

Committee Member: Dr. Michael Hadani, Ph.D.

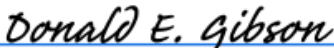
Date



04/23/2025

Committee Member: Dr. Nancy Lam, Ph.D.

Date

  
Donald E. Gibson (Apr 24, 2025 08:55 PDT)

04/24/2025

Dean of SEBA: Dr. Donald Gibson, Ph.D.

Date

## **Abstract**

Designers navigate uncertainty and opportunity as artificial intelligence (AI) reshapes the UX landscape. This dissertation presents an exploratory study investigating how UX designers perceive the increasing prominence of AI in their professional work. It also applies Albert Bandura's self-efficacy theory to organize the findings and examine how performance accomplishments, vicarious experiences, verbal persuasion, and physiological states influence their adaptation to AI. Findings indicate multiple perspectives in the sample interviewed: In this study, junior designers predominantly express anxiety over job security, while senior designers tend to embrace AI as a strategic partner that grants them a "seat at the table." Designers in this study primarily perceive AI not as a replacement but as a guide and collaborator. However, they acknowledge that AI will compress roles, requiring them to do more with fewer resources. There is also an agreement that designers need to adapt to the changes and design for the next generation by incorporating AI into their line of work. This dissertation contributes to the ongoing interest in AI's role by providing insights into designers' feelings about adapting AI into their work lives. The findings highlight the importance of fostering AI literacy and self-efficacy among UX professionals to ensure they can leverage AI effectively rather than feel displaced by it. By understanding how designers adapt to AI-driven changes, this research helps to lay the groundwork for further studies on cross-disciplinary collaboration, shifting job expectations, and AI's long-term impact on the UX design field.

## **Acknowledgment**

I want to express my deepest gratitude to my dissertation advisor, Dr. Michal Strahilevitz, whose invaluable guidance, insightful feedback, and unwavering support have been instrumental in shaping this dissertation. Her keen eye for clarity and rigor pushed me to refine my arguments, strengthen my theoretical grounding, and sharpen my analytical lens. Even when I was uncertain of my direction, she provided the patience and wisdom to help me navigate the complexities of academic research. Her dedication to excellence and her willingness to meticulously review multiple drafts enhanced the quality of this dissertation and transformed my approach from that of a practitioner to a researcher. I am deeply grateful for her mentorship, encouragement, and commitment to my academic growth.

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Completing this dissertation has been a demanding and transformative journey, and I could not have done it without the guidance and encouragement of my advisor and committee. Their generosity with their time, knowledge, and support has been invaluable throughout this process. Their belief in my work and their commitment to academic excellence have shaped this dissertation and my development as a scholar. For that, I am genuinely grateful.

## **Dedication**

This dissertation is dedicated to my parents, husband, and children, whose belief in me has strengthened and encouraged me throughout this journey. Your patience, love, and constant reassurance have lifted me in moments of doubt and exhaustion, reminding me why I embarked on this path.

To my family and friends, your boundless love, laughter, and resilience have kept me going through the challenges of this doctoral pursuit. You have given me purpose beyond words, and your understanding has made this achievement possible.

To my cohort mates, Brian Archibald, Cyrus Ghamghami, and Adam Stelly, who have been with me through countless drafts and iterations, thank you for your willingness to brainstorm, challenge my thinking, and help me refine my academic arguments. Your insights, encouragement, and belief in me have strengthened this dissertation and my confidence as a scholar.

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## **Introduction**

When I began my career as a User Experience (UX) designer at IBM, my team would spend at least three months gathering user feedback about specific problems, another six months designing features based on the users' feedback, and another year implementing the design with the engineering team. Through technological and operational advancements, the product development cycle is much more efficient using processes like Agile or Design Sprint, in which the team splits big projects into smaller, sizable chunks that are easier to design, develop, and implement. The design process is being transformed once again with the advancement of artificial intelligence (AI).

With AI becoming increasingly embedded and influential in every industry, it has revolutionized how we work and interact with technology. AI began as an experimental field with pioneers like George Boole (1815-1864), Allen Newell, and Herbert Simon, who founded the first artificial intelligence laboratory (Kumar, 2004). The AI phenomenon is not new, even though it has only recently gained momentum in management literature in the last few years (Bettis and Hu, 2018; Furman and Teodoridis, 2020; Raisch and Krakowski, 2021). During the mid-1950s, experts began discussions on the plausibility of artificially intelligent computer systems. Notably, management scholars such as Richard Cyert, James March, and Herbert Simon at the Carnegie School had already demonstrated interest in the potential impact of computer processing on organizations during this period (Haefner et al., 2021). Recent developments have made AI advance in ways that were impossible before (Choudhary et al., 2023). AI technology will impact how we work, as demonstrated by its estimated economic contribution of \$13 trillion by 2030, potentially boosting global GDP by 1.2 percent annually (Bughin, 2018).

AI's transformative potential has raised concerns about its impact on the workforce and organizational structures. While some view AI as an opportunity for creating a competitive edge through automation and work augmentation (Tschang and Almirall, 2021; Raisch and Krakowski, 2021), others perceive it as a threat that could potentially lead to job-specific obsolescence (Malik et al., 2019; Frey & Osborne, 2017). Automation suggests that machines should entirely assume control of human tasks by replacing a human worker with a robot that performs the same job. However, solely relying on automation might fuel a cycle of job displacement, widening socioeconomic inequality, or leading to the deskilling of the workforce. Instead, some scholars advocate using AI as an augmentation strategy in which humans and machines collaborate closely to perform tasks cooperatively, replacing time-consuming human activity with automated processes that improve accuracy, efficiency, and effectiveness (Langley & Simon, 1995).

As AI continues to revolutionize the professional landscape, the role of a UX designer is evolving in conjunction with these changes. Self-efficacy theory can help explain and better understand UX designers' perspectives and emotional responses amidst the changing landscape. In many other fields, integrating AI into the process brings an array of emotions for workers as they experience a mix of excitement and curiosity about the possibilities of new technology while simultaneously grappling with concerns, fears, and anxiety about the potential impact on their job security and professional identities (Park & Woo, 2022). With looming headlines like “300 million full-time jobs may be replaced by AI” (Vallance, 2023) and “46% of workers in advertising, marketing, business support, and logistics are worried AI will soon take over their jobs (Rosenbaum, 2023), it is crucial to investigate how designers are working with, or without, AI in the design process. Taking a time capsule of their feelings and attitudes during this

transformative time will allow us to identify ways to maximize the value of this emerging technology so that it can be evaluated for its relevance, significance, and contribution to the future of work in UX. It will also give us a better theory-driven understanding of how people respond to environmental changes in their professional lives through the lens of self-efficacy.

### **Self-Efficacy and Locus of Control**

Albert Bandura's theory on self-efficacy (Bandura, 1977) and Julian Rotter's locus of control (Rotter, 1966) have significantly influenced our understanding of motivation and perception of control over an individual's life events. Bandura's self-efficacy research shows that the belief in one's capability to execute a particular course of action is a critical determinant of human behavior. It is the basis for forming motivation, feelings, and personal achievements and is a crucial determinant of self-regulation (Steinbauer, 2018).

Locus of control, conceptualized by psychologist Julian Rotter in 1966, is a psychological construct representing a generalized personality trait. It defines how individuals attribute their successes and failures to internal determinants—such as personal effort and inherent ability—versus external circumstances—such as fate, luck, or the influence of powerful individuals (Rotter, 1966). It is used to illuminate how people perceive control over their life outcomes and has significant implications for understanding motivation, behavior, and psychological well-being. Research indicates that individuals with a strong internal locus of control tend to develop higher levels of self-efficacy, as they believe their actions influence outcomes (Skinner, 1996). In contrast, those with an external locus of control may experience lower self-efficacy, viewing

events as outside their influence, leading to a phenomenon known as learned helplessness (Seligman, 1972).

While both self-efficacy and locus of control offer valuable insights into human motivation and behavior, this study focuses on self-efficacy in relevance to how UX designers adapt to AI advancements and the impact on their role. Locus of control is a broader and more generalized personality trait that reflects how individuals attribute success or failure to internal or external factors. However, it does not explicitly account for how people develop confidence in performing tasks or acquiring new skills. Since this research examines how UX designers perceive their ability to learn and integrate AI into their workflows, self-efficacy provides a more precise framework. It captures the process of building competence through experience and accomplishment, observing others successfully perform a behavior, getting encouragement and feedback from others, and their emotional and psychological states—all factors that directly influence how designers navigate technological changes. By focusing on self-efficacy, this study explores the designer's feelings and emotions about AI through the four dimensions of self-efficacy rather than their broader worldview on control and external influences.

### **Self-Efficacy Theory**

Self-efficacy is the belief in one's ability to use those skills effectively in specific situations. It enhances performance by making people feel they have control of their destinies, increasing the difficulty of self-set goals, escalating the level of effort, and strengthening persistence (Bandura, 1977, 2015; Bandura & Locke, 2003). Self-efficacy influences our choices, the effort we expend, how long we must persevere in facing challenges, and how we cope with stress and anxiety. Individuals with high self-efficacy can increase strong self-

confidence regarding task achievement (Nusannas et al., 2020). For individuals with low self-efficacy, it can cause motivational problems, precisely when they believe they cannot succeed on specific tasks, impeding their achievement and creating self-fulfilling prophecies of failure (Margolis and McCabe, 2006).

There have been many studies on how self-efficacy can impact performance. In 2013, Sitzmann & Yeo conducted a meta-analysis investigating the relationship between self-efficacy and performance over time. Specifically, it examined whether self-efficacy primarily drives future performance or is a product of past performance. The authors analyzed data from 38 studies between 2001 and 2012, including 28 published and 10 unpublished works, with 5,414 participants. They found that past performance predicts self-efficacy more than future performance. It suggested that self-efficacy is primarily shaped by previous experiences and successes rather than being the primary force driving future achievement.

In contrast, those with low self-efficacy are less likely to engage in tasks that require specific skills and are less likely to persevere when faced with obstacles and challenges (Pajares, 2005). They are more inclined to avoid challenging tasks, give up quickly, and experience increased levels of stress and anxiety. Low self-efficacy is often associated with a negative cycle where individuals perceive challenges as threats rather than opportunities and can lead to reduced goal setting, minimal effort, and an increased likelihood of abandoning tasks when difficulties arise (Lent, Lopez, Brown, & Gore, 1996). However, strong self-efficacy beliefs can minimize the effects of other individual difference factors such as anxiety, physiological predisposition, and interest (Lent, Lopez, Brown, & Gore, 1996).

Self-efficacy also has a broader impact on career development and can be related to how UX designers perceive and adapt to integrating AI into their work. Self-efficacy beliefs are

specific judgments of capability that are goal-oriented and context-dependent, changing according to the task involved (Bandura, 2001.) They shape how individuals approach their professional growth, influencing their willingness to acquire new skills, adapt to changing environments, and engage in lifelong learning. Bergdahl & Sjöberg (2025) examined the significance of attitude and readiness in professional growth, particularly in integrating new technologies such as AI in education. The study explored teachers' readiness to use AI-driven chatbots in K-12 education. It used a mixed-methods approach with survey and poll data to investigate teachers' AI self-efficacy, attitudes, and perceived usefulness of AI in education. The findings suggest that teachers generally have positive attitudes towards AI's potential in education but their AI self-efficacy differs significantly depending on their previous experience with AI, how relevant they perceive it to be, and the amount of support they receive. The study emphasized the importance of AI self-efficacy in shaping user attitudes and adoption intentions.

This facet of self-efficacy is particularly pertinent in dynamic fields such as UX design, where rapid technological advancements, including the integration of AI, necessitate ongoing learning and adaptation. The self-efficacy theoretical framework provides a fresh perspective for analyzing and comprehending designers' behaviors and motivations with AI. Designers with strong self-efficacy are more inclined to view AI as a beneficial tool to enhance their work rather than a threat to their professional roles. This theory was selected after insights were gathered from the qualitative interviews.



## The Four Sources of Self-Efficacy

According to Bandura (1977), self-efficacy expectations come from four primary sources. These four sources influence an individual's perception of their efficacy, ultimately impacting their motivation and performance in facing challenges. Understanding these sources can be vital for this research as we learn about an individual's beliefs in their capabilities.

- 1) Performance of Accomplishments:** Mastery experiences, also known as performance accomplishments, serve as the most significant source of self-efficacy. When individuals confront a complex or unfamiliar task, they tend to reflect on their past performances in similar situations to assess their self-efficacy (Bandura, 1991, 1997; Wood & Bandura, 1989). These experiences provide direct evidence of one's capabilities, leading to more substantial and robust beliefs in self-efficacy. Previous successes enhance self-efficacy, while failures can undermine it. For instance, a student who consistently excels in math tests develops a strong belief in their mathematical abilities. In contrast, repeated failures, particularly early in skill development, can weaken one's self-efficacy (Schunk, 1995).

The impact of performance accomplishments is determined not only by the outcome itself but also by factors such as the perceived difficulty of the task, the effort invested, the amount of external support received, and the timing of successes and failures. Navigating obstacles through persistent effort bolsters self-efficacy more than easily attained successes (Carroll & Fox, 2017) and highlights the importance of viewing challenges as opportunities for growth and learning rather than indicators of inadequacy.

**2) Vicarious Experiences:** Observing someone who resembles us succeed can significantly enhance our confidence in our ability to master similar tasks. As supported studies on athletic training have shown, students found that watching a peer perform a skill improved self-efficacy (Carr & Volberding, 2014). The more we identify with the observed individual, the more profound the impact tends to be. This influence is powerful when tackling unfamiliar tasks or harboring doubts about our capabilities. Witnessing someone navigate challenges through persistent effort often has a more significant effect than seeing a flawless performance, as it emphasizes the value of perseverance and illustrates that success is achievable, even amidst initial struggles (Bandura, 1977). This process is a source of self-efficacy, shaping our perceptions of what we can develop and achieve.

**3) Verbal Persuasion:** Encouragement and positive feedback from credible sources can significantly enhance self-efficacy beliefs. When such feedback comes from respected mentors, teachers, or coaches, it can have a potent effect, especially when coupled with opportunities for the individual to achieve success (Bray-Clark & Bates, 2003.) However, when opportunities for skill development and successful performance are not accompanied by verbal persuasion, the benefits may be short-lived or even harmful, particularly if future experiences counter the positive reinforcement (Bandura, 1977). For persuasion to be most effective, it should be specific, concentrate on attainable goals, and emphasize the individual's strengths and potential for growth (Schunk, 2001). In Haro Soler's 2021 study, verbal persuasion positively influenced students' self-efficacy when

the comments were realistic and aligned with the student's actual abilities. The study concluded that a student-centered environment with a caring teaching approach is essential for verbal persuasion to effectively build self-efficacy in education (Haro Soler, 2021).

**4) Physiological and Emotional States:** Physical and emotional reactions—such as stress, anxiety, or excitement—serve as internal indicators that individuals utilize to evaluate their capabilities (Bandura, 1977). A pessimistic interpretation of these states can weaken self-efficacy, whereas an optimistic interpretation can strengthen it. Bodily responses like a racing heart or sweaty palms, alongside emotions such as anxiety or excitement, inform how we perceive challenging situations (Bicknell, 2021). An elevated state of arousal may be construed as a sign of being overwhelmed and lacking the skills to cope, thereby diminishing self-efficacy (Bandura, 2015). Conversely, the same physiological response could be interpreted as anticipatory excitement, indicating readiness and enhancing self-belief. Importantly, it is not the physiological or emotional state that matters but how it is interpreted, which ultimately influences self-efficacy. This interpretation is subjective and influenced by individual factors, past experiences, and the specific context. The failure to complete a task makes workers more stressed, especially if the task is unclear, the orders overlap, the schedule is conflicted, and demands for work results are too high (Semmer et al., 2015).

It is essential to acknowledge that the various sources of self-efficacy do not exert equal influence on individuals. Empirical evidence suggests that performance accomplishment

significantly impacts self-efficacy development, followed by vicarious experiences (Schunk, 1989). Verbal persuasion contributed somewhat, and physiological and emotional states exerted the least influence. This hierarchical ordering underscored the relative potency of each source in shaping an individual's confidence in their capabilities (Bandura, 1997).

Gaining insight into how sources of self-efficacy shape designers' attitudes toward AI offers valuable guidance for organizations seeking to support their teams with customized training, mentorship, and resources. It highlights the significance of cultivating a culture of learning and adaptation, empowering junior and senior designers to navigate the ever-evolving intersection of design and technology effectively.

## **Literature Review**

### **What is UX Design?**

To understand the impact of AI on UX design, we need to define the discipline first. UX design encompasses a multidisciplinary process to align user needs with business value. UX design takes a holistic approach to balancing pragmatic aspects with hedonic, non-task-related aspects of product possession and usage (Hassenzahl, 2006). The primary goal of a good UX design is to create a seamless and enjoyable experience when a user is interacting with a product. Users of such products have high expectations for experiences, and success is often linked to how designers understand and translate user requirements into corresponding functionality and appropriate aesthetics (Silva-Rodríguez et al., 2020). Innovative experiences are created when UX designers collaborate with interdisciplinary teams to combine research, strategy, and product development to build a product that fulfills the needs and expectations of users.

UX design is distinct from many other creative professions in that it is not solely artistic but also focuses on meeting the needs of end users (Unger and Chandler, 2023). It involves creating the visual appearance of a digital product and developing efficient navigation systems, labels, and content categorization to ensure that users can easily find and access the information they need (Hartson and Pyla, 2018).

UX design is inherently human-centric, relying on empathy to comprehend user needs and a profound understanding of human psychology to interpret user behavior. It requires creative and collaborative skills with a problem-solving and sense-making mindset (Oulasvirta et al., 2020). Successful UX design involves multiple iterations and is closely linked to how well designers understand and translate users' requirements into corresponding functionality and make technical and non-technical decisions. With the introduction of AI design tools, there is an opportunity to automate repetitive design tasks like wireframing and prototyping (Houde et al., 2022), facilitate brainstorming of design options, and accelerate the iteration process to develop more effective and engaging user experiences (Saadi and Yang, 2023).

AI has the potential to significantly transform the user experience design process and reshape the way UX designers approach their work. However, the successful adoption of this new technology is heavily contingent on employees' perspectives and attitudes about this technology. In the upcoming sections, I will review past research and case studies on employees' perceptions and attitudes about AI to identify trends and insights that could be applicable on a broader scale.

## **How Employees' Perceptions and Attitudes of AI Impact Adoption**

Employees' willingness to use AI in the workplace is influenced by their perception of the technology and their attitude toward AI can affect its adoption. As much as employees may want to rationally assess and evaluate the technology's capabilities based on their roles and responsibilities, they are influenced by their experiences and understanding of AI. How employees perceive and feel about AI plays a significant role in determining the integration and adoption of AI in the workplace.

Chiu et al. (2021) highlighted the importance of employees' attitudes in shaping their acceptance and usage of AI in the workplace. The authors surveyed 363 participants from various industries through a market research firm in Taiwan. They found that employees' initial opinions about AI significantly impacted their willingness to accept and use it. The study revealed that employees' perceptions of AI's operational and cognitive capabilities were linked to their emotional and mental attitudes toward AI, which implied that when employees believed that AI offered benefits such as reliability, flexibility, and integration with other systems, they were more likely to have positive attitudes at both the emotional and cognitive levels. Furthermore, the study suggested that knowledge about AI is a buffer against negativity toward it, and employees were more likely to have a balanced perspective if they comprehended the capabilities of AI. On the other hand, lack of understanding led to fears and anxieties, potentially resulting in resistance to adoption or intention to leave their position.

Using the same survey results from the market research firm from the above study, the second author conducted a broader analysis to find generalizable knowledge to predict employees' AI views. Zhu et al. (2021) analyzed the data to explore how employees reacted to the increasing presence of AI in the workplace, examining their perceptions of AI, rational and

emotional attitudes, and behavioral intentions. After the initial quantitative analysis, the study added qualitative insights by interviewing eight employees. The collected data identified four employee attitudes toward AI profiles: *Reticent*, *Intrepid*, *Skeptical*, and *Dissenter*.

- *AI Reticent* employees generally viewed AI positively, believing in its high operational and cognitive capabilities. However, they remained hesitant and anxious about potential negative impacts like job displacement.
- *AI Intrepid* employees held a positive outlook on AI's capabilities and were enthusiastic about the technology and perceived opportunities associated with it, leading to excitement rather than anxiety.
- *AI Skeptics* employees were less convinced about AI's capabilities and held a more neutral stance. They acknowledged AI's potential but remain unconvinced about its effectiveness and reliability.
- *AI Dissenters* employees held the most pessimistic view of AI and believed it offered little to no business value. They experienced strong negative emotions towards the technology, such as fear and anger, and resisted AI implementation in their workplace.

Contrary to expectations, an employee's age, gender, or job title were less significant in shaping their response to AI than their overall attitude towards technology, their understanding of AI's capabilities, and the nature of their work. Ultimately, the study emphasized that successful AI implementation depends on the employees' attitudes toward technology, so understanding their AI profile helps managers understand their perspectives and behaviors toward technology.

In addition to predicting AI adoption with profiles, a manager's leadership style also plays a crucial role in AI adoption. Kolbjørnsrud et al. (2017) examined the complexities of integrating AI into organizations, highlighting the significant variations in managers' receptiveness to AI adoption. Through Accenture's survey of 1,770 managers in 14 countries and interviews with 37 senior executives, the authors discovered that senior managers in this study were more at ease with AI than their junior counterparts. They suggested successful AI adoption hinged on a tailored approach, recognizing that managers at different levels and diverse geographical locations hold varying degrees of understanding and acceptance of AI. Furthermore, geographical disparities were evident, with managers in emerging economies displaying a greater comfort with AI than those in developed nations. It proposes that managers in emerging economies may have been eager to adopt forward-looking tools to achieve global best practices and leapfrog the competition. The authors concluded that a one-size-fits-all approach to AI implementation was unlikely to succeed and advocated for strategies that acknowledged and addressed these variations in managerial attitudes and readiness.

Probing deeper into the employees' experience in the role, Wang et al. (2023) discovered that junior employees were able to benefit more from AI than employees with a longer tenure. The authors implemented a machine-learning AI for a medical coding system in a publicly traded U.S. healthcare company. The study assessed the varied impact of AI on productivity by measuring the volume of tasks performed by the employees and their tenure on the job. The research found that junior employees with specific tasks experienced more significant gains from AI, while senior workers showed lower productivity gains. A follow-up qualitative field study and laboratory experiment revealed that senior medical coders tended to trust AI less. They were more likely to detect imperfections in the AI output and resist adoption, hindering their capacity



to benefit from the potential productivity gains associated with AI use. This may be due to the high degree of accuracy and adherence to established guidelines required in their work. Medical coders analyze medical records and assign standardized codes for diagnoses and procedures, which diminished their tolerance for ambiguity. These findings highlighted the importance of considering different dimensions of experience when integrating AI into the workplace. The authors emphasized the need to address trust barriers to maximize the benefits of AI across all levels of seniority.

The above studies emphasized the crucial role of employee attitudes and experiences in shaping the acceptance and adoption of AI in the workplace. It showed that individual experiences, emotional attitudes, and cognitive understanding significantly influenced the willingness to use AI in the workplace. The successful deployment of AI is not only about the technology itself but also about understanding and managing diverse employee perspectives and behaviors. Additionally, the varying receptiveness of managers to AI adoption was highlighted, suggesting the need for tailored approaches that acknowledge geographic, hierarchical, and tenure-related differences.

This section revealed some studies on how employees' perceptions of AI influence adoption. The subsequent section will investigate how employees' perception influences their career behaviors.

### **How Employees' Perceptions of AI Impact Career Attitudes and Behavior**

Introducing new technologies and automation into the workplace often affects employees, leading to significant shifts in the skills needed for success. While automation has sometimes benefited society by increasing efficiencies and work productivity, segments of

employees have experienced disruptions in their careers. AI experts Müller and Bostrom (2016) predicted that AI systems will likely reach overall human ability by 2075 and that further progress of AI toward superintelligence may not be good for humanity. AI integration in organizations can create higher-skilled jobs, but it also can potentially displace millions of jobs in the economy (Dwivedi et al., 2021). Employees now fear that AI may reduce the demand for humans because AI may become more proficient in tasks traditionally performed by humans (Makridakis, 2017; Shneiderman, 2022) and can create a sense of insecurity and apprehension about the future of their careers. While some individuals have been able to adapt to these changes and improve their marketable skills, many employees feel anxious about job displacement, job insecurity, and reduced job satisfaction. As AI takes over decision-making in the workplace, employees may lose the skills necessary for their jobs.

A study by Danielsen (2023) explored the consequences of AI implementation on human decision-making processes in the workplace in the financial sector. The case study examined the implementation of an AI system called “Clever Loan” in a leading German bank, which was implemented to replace employees’ decisions for loan approvals. The researchers found that outsourcing employees’ decision-making to AI could alter employees’ sense of responsibility. Employees felt less accountable for the outcomes, potentially negatively impacting their motivation and job engagement. While the organization aimed to increase profit and decrease default rates, the employees reported a loss of purpose and connection, contributing to their disengagement from work as they experienced a diminished sense of purpose.

Koo et al. (2021) also conducted a similar study in the hotel industry to analyze the perception of AI and its effects on job insecurity, job engagement, and turnover intention. In a mixed-methods research design with surveys followed by a qualitative case study, the

researchers found a significant association between job insecurity and job engagement when the perception of AI was used. They revealed that employees who felt threatened by AI were less likely to be engaged in their work, regardless of whether they were in a managerial or non-managerial role. Additionally, the survey found a correlation between job insecurity and turnover intention, with employees more inclined to contemplate leaving their jobs if they were concerned about AI replacing their roles. However, the research also suggested that higher job engagement was a mitigating factor for job insecurity, demonstrating that more engaged employees were likely to remain in their positions despite concerns about AI.

It is argued that AI can help improve efficiency and productivity in the workplace. However, as a company continues to invest in AI, it becomes essential to consider how these decisions may impact employees in the long term. Presbitero and Teng-Calleja (2022) set out to find out how AI's continuing integration and growth in the workplace can influence employees' career attitudes and behaviors with a longitudinal study. With the help of the HR director from an international call center in the Philippines, the authors sent four surveys to call center agents who had recently integrated AI into their operations within a month. The study showed that the perception that AI is threatening jobs was positively and significantly related to the employees' career exploration behavior, prompting employees to explore alternative career paths and consider leaving their current positions due to job insecurity. Feelings of job insecurity were also significantly associated with heightened experiences of psychological distress like anxiety, depression, and perceived stress among employees. The researchers presumed that career exploration behavior was an adaptive mechanism which enabled an employee to reduce the heightened psychological distress from job insecurity. Employees' intention to leave and

search for other career opportunities laid on the expectation that other careers would have been better because they caused less psychological distress.

In a similar study, Brougham and Haar (2018) explored how employees viewed their future jobs and careers in technological advancement. Using online and paper-based surveys, the researchers collected data from the service sector in New Zealand to measure employee awareness of STARA (Smart Technology, Artificial Intelligence, Robotics, and Algorithms) and other factors like organizational commitment, career satisfaction, turnover intention, depression, and cynicism. The survey also had an open-ended question to capture participants' comments on how STARA affected their current jobs and how it might have impacted their future career prospects. The results from the study showed that when employees were more aware of STARA and its application to their jobs and careers, they were more likely to have lower organizational commitment and career satisfaction. In addition, employees with a higher perception of STARA were likely to have had higher adverse effects on turnover intentions, depression, and cynicism. The qualitative finding further explained that traditional career planning was no longer the norm, and employees were more open to the idea of boundaryless employment since the job or industry might disappear due to technological advancement.

Employees' attitudes and behaviors toward their careers are significantly influenced by their perception of AI. The introduction of AI and automation has raised concerns about job satisfaction, engagement, security, and intention to leave among employees. Studies have revealed that employees who view AI as threatening are less committed and more inclined to consider other career options. Moreover, the fear of AI replacing human roles has been linked to higher levels of psychological stress. While AI can improve workplace efficiency, organizations

may wish to address the effects of these technological advancements on employee emotions and career growth to improve employees' attitudes and behaviors.

This section has featured several studies examining the influence of employees' perceptions of AI on their career choices and behaviors. The subsequent section will dive deeper into the impact of employee perceptions on creativity, a fundamental skill for a UX designer.

### **How Employees' Perceptions of AI Impact Creativity**

Employees' perceptions and attitudes can significantly impact AI adoption and career behavior. But how do the perceptions of AI affect an employee's creativity, especially for UX designers? UX is a discipline that blends art and science in its practice. UX designers analyze user behavior and motivation to enhance product usability while heavily relying on creative thinking to ensure that business and technical requirements align with user needs. For designers, introducing AI in the creative process may elicit excitement and curiosity about its possibilities, and some studies have predicted that humans will shift towards more creative and cognitive roles to support AI technologies (Jönsson & Svensson, 2016).

To understand whether AI tools can enhance creativity, Jia et al. (2024) investigated the impact of AI assistance on employee creativity and how this effect might differ based on employee skill levels in a telemarketing company. First, a field experiment was implemented where one group of employees received AI assistance in generating sales leads while a control group did not. The researchers then measured the creativity of employees' responses to customer questions and their overall sales performance. Second, semi-structured interviews were conducted with employees from both groups to gain qualitative insights into their experiences

and perceptions of AI assistance. The findings revealed that AI assistance boosted employee creativity in higher-skilled employees because they could leverage the shift, leading to positive emotions that foster creativity, more creative sales interactions, and, ultimately, higher sales. Conversely, lower-skilled employees struggled to adapt, experienced negative emotions, and showed limited improvement in their creative output.

Looking deeper into employees' perceptions of AI to the UX field, one empirical study was found as of June 2024. A conference proceeding by Li et al. (2024) examined UX designers' perceptions of AI and the evolving technological landscape by conducting a qualitative study, interviewing 20 UX professionals representing a diverse range of backgrounds and varying levels of experience. The researchers aimed to understand how UX designers viewed the role of AI tools in their workflows and processes while identifying the opportunities and challenges presented. This study investigated how AI tools were incorporated into existing design processes and their perceived impact on the overall quality and efficiency of UX design.

The findings from the study revealed a generally optimistic outlook among experienced UX designers. Experienced designers viewed AI as an assistant, acknowledging the potential of these AI tools to automate repetitive tasks, enhance productivity, and even foster creative design exploration. The authors emphasized the importance of human skills, highlighting the designer's role in guiding AI output and ensuring its alignment with user needs. However, concerns were raised regarding the potential negative impacts on designers. If junior designers had become overly reliant on AI to generate design solutions or perform tasks fundamental to developing core UX skills, it could hinder their learning and development as designers.

While AI can be a valuable tool for UX designers, it is crucial for them to use it appropriately and strike a balance. AI tools should be used strategically to enhance the creative

workflow while allowing designers to actively develop a strong foundation in fundamental UX principles and practices. Teresa Amabile, a leading scholar and researcher on creativity, wrote a guidepost article in 2020 stating that “creativity and AI [research] will be one of the most exciting—and necessary—developments of research and practice on creativity in the 21st century.”

As much as the conference paper by Li et al. (2024) examined UX designers’ perceptions of AI and the tools associated with 20 UX professionals, they did not discuss the emotional impact this technology has on a UX designer. My dissertation investigates the thoughts and feelings of UX designers, how AI has, and will, change their role, and the risks and benefits of AI technology on their careers.

### **Gaps in Literature and Research Questions**

These studies underscored the impact of AI integration based on employees' perceptions and attitudes towards AI adoption in the workplace, its effects on their career behaviors, and its impact on the creative role and process. Prior work showed that some employees embraced AI as a valuable tool that complemented human creativity and enhanced work productivity. In contrast, others harbored concerns about the potential negative impacts, such as job insecurity, job satisfaction, job displacement, and skill degradation. While numerous studies had examined employee perceptions of AI adoption and its impact on career trajectories, limited research has been found on AI’s influence on UX designers.

Understanding the hopes, concerns, and emotional impact of AI on UX designers will help uncover AI's potential in fostering creativity, thereby reshaping the role while enhancing job satisfaction and engagement. Knowing the varying perceptions and emotional responses to AI

will help managers and organizations understand how AI can be incorporated into the design process so that designers are equipped with the necessary skills and knowledge to work alongside AI tools effectively.

This dissertation explores the thoughts and feelings that come up for UX designers when incorporating AI into their professional work. I also seek to gain insights into how AI has impacted the role of UX designers in recent years. Finally, I will explore the perceived risks and benefits designers associate with AI. Specifically, the following research questions (RQs) will be explored:

**RQ1:** What themes emerge when UX designers share their thoughts and feelings about AI in the workplace?

**RQ2:** How has AI changed designers' roles in recent years?

**RQ3:** What do designers perceive as potential risks and benefits of AI?

These perspectives will facilitate a better understanding of the emotional impact of AI in the role of a UX designer. Ideally, the findings will help identify better strategies for encouraging designers to incorporate AI in the workplace.

## **Recruitment Method, Participants, Data Analysis Methods**

### **Recruitment Method**

Based on the exploratory nature of the research questions in this study, a qualitative research design grounded in phenomenology is chosen. Phenomenology is a philosophical approach that seeks to understand the essence of lived experience, emphasizing the subjective meaning individuals ascribe to their experiences (Moustakas, 1994). This method serves the



purpose of gaining an in-depth understanding and uncovering nuanced findings compared to other methods of collecting data. It perceives reality from human perspectives and gathers data that cannot be easily measured (Creswell & Creswell, 2018). It focuses on the participant's meaning rather than their previous opinions and knowledge or the information from other literature. The inductive method is central to this study, aiming to derive themes and patterns directly from the data without imposing preconceived theoretical frameworks (Creswell & Creswell, 2018). This approach aligns with the exploratory nature of the research, allowing for a deeper understanding of the phenomenon before aligning it with existing theories or potentially generating new ones.

For this study, I have conducted 20 qualitative, semi-structured interviews with UX designers of various backgrounds and experiences. Interviews were conducted with designers working or have worked in the last 6 months, ranging from startups to large corporations. Half (10) of the designers were junior designers with 1-4 years of experience, and the other half were senior designers with more than five years of experience. The two populations are selected based on a study by Ahmed and Wallace (2004), in which they identified noticeable differences in the behaviors of new entrants to the engineering design profession compared to more experienced colleagues. The juniors utilized a "trial and error" approach, generating and implementing design modifications, evaluating them, and then repeating the process through multiple iterations.

In contrast, experienced design engineers were observed to conduct a preliminary evaluation of their tentative decisions before implementing them and making a final assessment. They carefully considered whether it was worthwhile to proceed further into the implementation stage of a design decision. Therefore, I anticipated similar distinctions and behaviors between the

groups as I interviewed the junior and senior designers to learn about their practices and explore their thoughts and feelings regarding AI in their roles and their perspectives on the future of UX.

One-on-one interviews on Zoom and unmoderated interviews were conducted via User Testing. Each session lasted between 15-45 minutes. The interviews were recorded for transcription and analysis. As part of the process, participants were assigned pseudonyms, such as J1-10 for junior designers and S1-10 for senior designers. Additionally, the following details were collected from the participants: name, email address, employment status, years of professional experience, company size, and industry, so characteristics or patterns may be identified.

This study utilized a phenomenological qualitative research design to explore the essence of human experiences and understand the importance individuals attribute to them. It is used to reveal the underlying themes of these experiences without imposing predefined theories or interpretations to emphasize the intricate relationship between human emotions and technological integration among UX designers. The qualitative interviews with semi-structured questions and unmonitored surveys, were comprised of four sections. The first section was initiated with general inquiries to acquire information about participants' backgrounds, work experiences, involvement with AI, and their perspectives on AI's current and future state. The second section explored the impact of AI on a designer's role, exploring the specific ways in which AI has influenced a designer's work. The third section focused on understanding AI's benefits and drawbacks. Lastly, the fourth section gathered designers' thoughts on the future of UX with concluding questions. Interview questions can be found in Appendix A.

**Stage 1:** Introduction was given by the interview facilitator to ensure participants understood the purpose of the interview and had read, comprehended, and consented to the interview and recording.

**Stage 2:** Questions on the designer's current role and responsibilities at their current job and their feelings toward AI.

**Stage 3:** Questions on how AI has impacted a designer's role.

**Stage 4:** Questions on the potential risks and benefits of integrating AI into the design process.

**Stage 5:** Concluding questions.

After the session, recordings were transcribed using a user research transcription and research repository service called Loop Panel. The recordings will be deleted 12 months after the project's completion. To safeguard data privacy, I ensured that participants' responses were not attributed to their names and pseudonyms were used in the results section instead.

## **Participants**

Designers of various levels were recruited through my professional social network on LinkedIn. Since I have been in the UX field for more than 20 years and have worked in enterprise to mid-size tech and non-tech companies, I have many professional connections that fit the criteria of this study. Furthermore, I teach UX design in an undergraduate and graduate UX program, where I have access to many UX designers with various degrees of experience. Participants were rewarded with a \$25 Amazon gift card to thank them for participating in this study.

Ten (10) junior and ten (10) senior designers were invited to participate in this study. Table 1 below lists the title, years of experience, company size, and industry of the participants. The participants in this study represented a diverse range of design professionals, primarily UX and product designers, working across various industries and company sizes. The sample included both junior (J1–J10) and senior (S1–S10) designers, with experience levels ranging from 1 to 25 years. The junior participants, who have between 1 and 4 years of experience, hold roles such as UX Designer, Web Designer, and Program Manager, working in industries including entertainment, automotive, finance, government, and hospitality. They are employed at companies ranging from small design firms to large enterprises with over 10,000 employees.

The senior participants, with 5 to 25 years of experience, have more advanced roles such as Senior UX Designer, UX Strategist, and VP of Product Design. They work across industries including consulting, healthcare, real estate, and digital marketing, with some operating as independent professionals while others are employed by large corporations. The inclusion of participants from both junior and senior levels provided a comprehensive perspective on how AI can influence designers' roles in different industries, capturing the anxieties of early-career designers and the strategic insights of more experienced professionals.

| #   | Method      | Years of experience | Title                          | Company Size | Industry          |
|-----|-------------|---------------------|--------------------------------|--------------|-------------------|
| J1  | Interview   | 4 years             | UX Designer & Program Manager  | 10,000+      | Entertainment     |
| J2  | Interview   | 2 years             | Web Designer                   | 51-200       | Oil and Energy    |
| J3  | Interview   | 2 years             | UX Designer                    | 1-50         | Design            |
| J4  | Interview   | 1-2 years           | UX Designer                    | 1001-5000    | Financial         |
| J5  | Unmoderated | 1-2 years           | UX Designer                    | 10,000+      | Automotive        |
| J6  | Unmoderated | 3-4 years           | UX Designer                    | 5001-10,000  | Design            |
| J7  | Unmoderated | 3-4 years           | UX Designer                    | 10,000+      | Government        |
| J8  | Unmoderated | 3-4 years           | UX Designer                    | 501-1000     | Hospitality       |
| J9  | Unmoderated | 1-2 years           | UX Designer                    | 10,000+      | Design            |
| J10 | Unmoderated | 3-4 years           | UX Designer                    | 1            | Design            |
| S1  | Interview   | 25 years            | Sr UX Designer                 | 1            | Entertainment     |
| S2  | Interview   | 10 years            | UX Designer/Digital Strategist | 1            | Digital Marketing |
| S3  | Interview   | 8 years             | VP Product Design              | 10,000+      | Consulting        |
| S4  | Interview   | 7 years             | Sr Product Designer            | 1-50         | Design            |
| S5  | Interview   | 7 years             | Sr Experience Designer         | 501-1000     | Design            |
| S6  | Interview   | 15+ years           | Sr UX Strategist               | 501-1000     | Healthcare        |
| S7  | Interview   | 12 years            | Sr Product Designer            | 1-50         | Real Estate       |
| S8  | Unmoderated | 5 years             | Product Designer               | 1-50         | Design            |
| S9  | Unmoderated | 5 years             | Product Designer               | 1-50         | Mining            |
| S10 | Unmoderated | 5 years             | Product Designer               | N/A          | Design            |

*Table 1: Summary of Participants Information*

## Data Analysis Methods

According to Glaser and Strauss (1967), the constant comparative method of qualitative analysis is the analysis of data within a grounded theory framework. This methodological approach entails a systematic process governed by precise coding and analytical procedures conducted by clearly defined steps.

**Phase 1:** The analysis of each participant's interview started with the first step of constant comparative analysis, which involved coding each incident found within the data. These incidents represented key concepts that emerged from the raw data. The initial codes were compared to develop categories that stand out in the findings. This comparative process fosters

the emergence of the theoretical properties associated with each category (Glaser & Strauss, 1967). In alignment with the principles of grounded theory, data analysis began concurrently with the data collection process. Transcribed interviews were initially coded and then later reviewed through focused coding, which entailed comparing the initial codes with one another to identify overarching categories.

**Phase 2:** The subsequent level of data analysis employed a constant comparative analysis approach, which integrated categories and their properties through the ongoing comparison of categories to identify themes (Glaser & Strauss, 1967). This integration involved analyzing each category with the others until thematic insights emerged that captured the experiences relevant to the research questions. Emerging themes, supported by direct quotes from both junior and senior designers, were presented first in the findings. While discrepancies between the themes articulated by junior and senior designers were noted, these differences strengthened the findings, as examining both perspectives was central to this research.

**Phase 3:** After completing all prior data analysis steps, patterns and themes about their thoughts and feelings about AI were identified. Additionally, the most dominant themes of the study were reported and discussed.

### **Reliability and Validity**

Qualitative studies are characterized by the researcher's interpretative nature and various data collection methods. However, researchers face numerous challenges regarding reliability and validity. Consequently, it is essential to implement measures that safeguard against these issues to ensure that the conclusions drawn from this research method are robust. A primary

concern arises from substantial data gathered from participants through interviews, which are often based on spoken words. The researcher's interpretation of participants' words introduces a notable threat to the reliability and validity of qualitative research due to potential researcher bias related to the interpretation of participants' words (Creswell, 2008). To mitigate the impact of subjectivity or researcher bias, I actively monitored my perspectives, enhancing the trustworthiness of the study's findings. While monitoring subjectivity does not eliminate it from qualitative research, it bolsters the reliability and validity of the conclusions drawn from the data by ensuring that they are acknowledged during the analysis (Glesne, 2006).

## **Findings**

The findings of this study provide a nuanced perspective on AI's influence on the role of UX designers. Designers shared their experiences through in-depth semi-structured interviews and unmonitored surveys, emphasizing AI's function as a guide and collaborative partner. They also expressed their feelings regarding their job security in an AI-driven industry. Finally, we looked at the evolving relationship between AI technology and the design process, highlighting how designers adapt to and navigate AI's impacts in their work and career trajectories.

### **AI as a Guide and a Collaborative Partner**

AI has quickly become an invaluable tool for many designers, not only as a productivity booster but also as a trusted guide and collaborative partner throughout the design process. Designers are increasingly using AI to support decision-making, ideation, and prototyping, with many seeing it as a tool that enhances their creative process rather than replacing it.

One of the most common ways AI assists is by guiding designers through the complex decision-making process. Designers often use AI as a checkpoint for their design flows and ideas. For instance, S7 noted,

*“It either gave me a clue of the direction I need to go, or I use it as like a reassurance...Yes, this is right, this is the way I need to be going when addressing this kind of flow. I use AI tools like ChatGPT as a sounding board to think through design problems and I have found that to be very, very helpful in the design world.” (S7)*

This quote illustrates how AI can provide guidance and reassurance, helping designers validate their approaches and confirm they are on the right path. AI can serve as a sounding board, allowing designers to potentially brainstorm and work through challenges. This reflects the concept of self-efficacy, as AI's role in validation and reassurance can enhance designers' confidence in their decision-making. This usage reflects AI's increasingly common role in ideation, facilitating a back-and-forth dialogue like brainstorming with a colleague. Another senior designer echoed this sentiment, explaining that *“sometimes everyone else is busy, or I don't want to keep bothering someone with specific questions, so I use AI as a partner to ideate and get different opinions.” (S5)* By enabling real-time feedback and idea generation, AI can allow designers to overcome creative blocks and refine their concepts more quickly, which in turn strengthens their belief in their ability to execute complex design tasks successfully, a key component of self-efficacy.

In addition to its role as a sounding board, AI has emerged as a potential collaborative partner in more structured ways, especially in providing iterative feedback. S7 shared, *“Using AI as a guiding stone to help give me feedback in real-time has been really helpful. It's like having someone ask, ‘Have you thought about it this way?’ or suggesting a different approach.” (S7)*



The immediacy of AI feedback allows designers to make informed decisions in real-time, allowing for a more agile workflow where design solutions can be refined and improved quickly.

The collaborative nature of AI is particularly evident in how research participants describe the ways it facilitates idea validation and inspiration. AI tools help designers quickly validate their intuitions or explore alternative approaches. As S5 explained, *“I use it to double-check my intuition, to see what other people are doing for similar tasks or solutions. It’s a shortcut, but I always read, edit, and tweak the output.”* (S5) This type of collaboration gives designers the confidence to pursue ideas they might have hesitated to explore otherwise. By reinforcing their ability to generate and refine creative solutions, AI plays a role in strengthening designers’ self-efficacy. It is also a means to keep the momentum going in fast-paced design environments.

Interview transcripts also suggested ways that AI can support designers when they need guidance on specific tasks or are unsure of the next steps. S6 described how AI helped with distilling complex information in moments of confusion. *“When I didn’t quite understand something or needed more clarity, AI helped me get a clearer picture and distill the information.”* (S6) This suggests that when providing more clarity or offering a fresh perspective on an idea, AI can function as a mentor and/or a collaborator, thus enhancing the designer’s process.

Furthermore, AI can help designers synthesize and summarize information. S8 said, *“You can use it to summarize your sticky notes. I was participating in a lot of user research activities, so I used the summarize feature in Dovetail...That’s been really helpful for me.”* (S8)

Participants noted that AI can help summarize a large amount of information, identify emerging

trends and patterns, and provide designers with insights that can help refine their designs. AI has shown to help “*create themes and trends out of usability studies*” (S6), simplifying what would otherwise be a time-consuming task.

Beyond these functional uses, AI’s ability to improve team communication and streamline mundane tasks is another theme that emerged from the interviews with designers. Many participants reported using AI to annotate emails and clarify complex concepts. One designer described how AI simplified their workflow: “*I now write all of my annotations on the new chatbot my company has created... it has really simplified all of my wireframes and made them a lot easier for developers.*” (J4). This suggests that, in some cases, AI is not just a tool for solo work; it can also enhance collaboration by reducing friction in communication and making design processes more efficient. AI can enhance designers' self-efficacy by providing verbal persuasion—one of the four sources of self-efficacy.

The availability of AI as a potential guide and collaborative partner was also mentioned as a benefit during iterative design. As one participant explained, “*Small changes, a lot of times, lead to really great results. I hope that AI will enable us to make more rounds of changes and even greater designs and experiences.*” (J2) This iterative feedback loop, facilitated by AI, allows designers to refine their work more frequently and efficiently, ultimately leading to better outcomes.

While the interview transcripts suggested that some UX designers see AI as a collaborative partner, it is essential to note that the designers interviewed, for the most part, still saw it as a tool to complement and not replace their skills. J8 emphasized,

*"I think it has the potential to take all of that repetitive and annoying work off of our plates so that we have time to do what people do best that AI cannot recreate, which is creativity and empathy and connection and all of those things." (J8)*

Designers still wish to incorporate their expertise and creativity, using AI to augment, rather than automate, their role. This collaborative approach could potentially enable designers to explore new possibilities and push the boundaries of their craft while maintaining their unique artistic vision.

Based on the transcripts, AI appeared to provide direction, feedback, and ideation support while improving efficiency and communication. However, the designers who participated in this study, for the most part, viewed AI as a partner that offered guidance and enhanced their decision-making, but ultimately still relied on human creativity and insight to drive innovative work.

In the upcoming section, I explore the nuanced perspectives of designers regarding job displacement, separated by junior and senior designers' insights and concerns, as well as the ways they view the balance between leveraging AI's capabilities and preserving the essence of human creativity in their work.

### **Job Security: Will AI Replace UX Designers?**

As AI continues to reshape the design world, designers expressed mixed emotions and varying levels of concern, particularly around the notion that AI could replace them. Some senior designers remained confident that AI would not directly threaten their roles. In contrast, others, particularly junior designers, expressed a blend of fear and uncertainty about their future in an AI-driven industry.

## Perspectives of Senior Designers

For many senior designers, AI is viewed not as a replacement but as a tool—an aid that enhanced their workflow rather than diminishing their job security. S5 stated, *“I don’t think it’s under threat. AI might be useful for more automated tasks, like language recognition, but design is a deeply human process. We still need someone to direct AI and tell it what to do”* (S5). This sentiment echoed a belief that AI can improve efficiency but cannot replicate the nuanced decision-making and creativity inherent in design work.

Another senior designer reiterated this point, explaining that even with the potential for increased AI involvement, *“there will always be a need for someone to facilitate AI, serve as an intermediary, and refine outputs. So, I’m not worried about AI taking my job. It’s more about learning to use it effectively”* (S7). This perspective aligns with Bandura’s (1977) theory of self-efficacy, which suggested that those with a strong belief in their ability to manage new challenges are more likely to view technological change as an opportunity rather than a threat. For these experienced professionals, the key to adapting and enhancing skills is to manage AI tools, such as interpreting, refining prompts, and understanding AI-generated outputs so that *“you can make sure the output of the AI is correct.”* (S5)

There was also an acknowledgment of the limitations of AI, with some senior designers asserting that while AI can aid in gathering information or generating ideas, it still cannot replicate the full scope of a designer’s role. *“AI hasn’t changed my work yet. But I can see how it might help with the research phase, almost like an advanced Google search.”* (S4) However, this

same designer also shared his view that aspects of design—such as critical thinking, problem-solving, and user empathy—may not be in AI’s grasp currently, thus being confident that human designers would still be needed in the near future.

*“When we're operating in a visual space, I think there's going to be gaps that at least as far ahead as I can see it will not be able to fill. [In] the creative problem space, I think that's where [AI is] really going to struggle. I can prompt it really, really well, but can it talk to people and understand unique situations and synthesize that down into really digestible problems and plans of action that are specific to what we're working on?” (S4)*

Despite these reassurances, there were hints of caution among some senior designers who felt uneasy about how AI is being adopted in the industry. One designer remarked, *“When AI first came on the scene, there was this hype about it replacing people...There is going to be such a dramatic shift in how everything is produced, that it just creates something we can’t really foresee... I think it’s less about AI killing jobs now, but it will definitely change how everything is produced.”* (S1) This fear concerns the shifts in the design landscape, where industries and roles could evolve in how things are produced. This uncertainty reflects the role of emotional states in self-efficacy development (Bandura, 1997). While senior designers demonstrated confidence in their ability to adapt, they also recognized that major shifts in industry practices can create stress and ambiguity, requiring continuous learning and adaptation to maintain their sense of control over their careers.

### **Perspectives of Junior Designers**

In contrast, junior designers, for the most part, expressed a more immediate concern about their job security in the face of AI. One junior designer shared, *“Yeah, it’s definitely scary. I feel like it’s going to change things, but I don’t know how.”* (J1) This expression of uncertainty

is common among those new to the field as they are still learning the ropes and fear that AI may reduce the demand for basic design skills. Another junior designer echoed concerns, saying, *“I think it will be harder for junior designers to break into the job market as AI continues to evolve. There’ll be less need for basic talent and more demand for niche and advanced skills.”* (J8) This perception may have stemmed from the fear that entry-level design tasks like creating basic layouts or prototypes may soon be automated by AI, leaving fewer opportunities for newcomers to enter the field. From a self-efficacy perspective, junior designers may experience lower self-efficacy, making them more susceptible to self-doubt and struggle to develop the confidence needed to adapt. However, as Bandura (1977) noted, self-efficacy can be enhanced through performance accomplishment and vicarious experiences. With time and exposure to AI tools, junior designers may develop the belief that they can successfully integrate these technologies into their workflows.

The fear of quite a few of the junior designers who participated in this study seems to be of AI replacing jobs and the implications for the broader design ecosystem. One junior designer noted, *“I worry that AI could make certain parts of our work redundant. What if AI can generate interfaces and prototypes faster than I can?”* (J7) This fear highlighted concerns that some junior designers, particularly those with low self-efficacy who are still developing their skills, may feel like they could be left behind as AI continues to advance.

However, some junior designers maintained a more optimistic outlook. One designer confidently stated, *“If you’re a good designer, you shouldn’t worry about AI taking your job. Design isn’t just about making pretty containers or buttons; it’s about flow and user experience. AI can’t replace the thoughtfulness and intuition that a designer brings to the table.”* (J4) This

reflected a belief that, while AI may shift the nature of design work, there will always be a place for human designers who focus on the deeper aspects of user experience. Like senior designers, cautious junior designers understand that AI could change their roles but not necessarily eliminate them. J4 also noted, *“It’s not that AI is going to steal my job. It’s more about it changing my job. As AI evolves, I’ll have to learn how to adapt and use it for my advantage.”*

(J4) This sentiment illustrated the growing recognition by some designers that AI can be a tool to be leveraged rather than feared and that those who embrace it and learn to work alongside it will remain relevant in the field. This perspective reflected a sense of high self-efficacy, as some designers expressed confidence in their ability to adapt, develop new skills, and integrate AI into their workflows to maintain their professional relevance.

### **Perspectives on Changes to the Role**

As both senior and junior designers reflected on AI’s role in the future of design, a common theme emerged: UX designers who participated in this study believed that AI will not replace designers but will undoubtedly transform their roles. They describe a future where AI serves as an augmentation model to enhance and accelerate specific tasks, allowing human designers to focus on the core elements of creativity, strategy, and empathy.

One senior designer said, *“I don’t think AI will take over the design profession. It will be a tool that helps us work faster and more efficiently. However, you still need someone to guide the process, make critical decisions, and ensure that the user’s needs are at the heart of the design.”* (S10) For junior designers, this perspective may provide some hope that their role will evolve rather than disappear. There was an expressed sense among many that as long as they remain adaptable and focus on the unique value that human designers bring to the table, they can

continue to thrive in an increasingly AI-driven industry. There appeared to be a strong sense that AI will continue to change how design work is done, with designers hoping and expecting that they will continue to play a crucial role in guiding the process with the help of AI providing strategic direction and ensuring that user-centric principles remain at the core of design practice. S1 emphasized the belief that, while AI will be a powerful tool, designers who can adapt, use AI effectively, and apply their unique human insights will remain irreplaceable. *"AI is not going to replace your job. But the people that use AI will."* (S1). This perspective reflected a high sense of self-efficacy as this designer expresses confidence in their ability to adapt, integrate AI into their workflows, and continue to play a crucial role in the design process.

### **Look Beyond the 8-Ball: What Do the Designers in this Study Think the Future of UX Would Be Like?**

As the role of UX designers continues to evolve in response to advancements in AI technology, many designers have expressed a belief that the future will blend human creativity and intuition with AI's efficiency and computational power. As AI streamlines workflows and automates specific tasks, designers noted that they are being asked to do more with fewer resources, adapt to new responsibilities, and anticipate the needs of future users with AI.

#### **Compression: Do More with Fewer Designers**



The efficiency brought by AI is expected to compress roles and team size, leading to more demands on designers to balance both technical proficiency and creative insight with fewer designers. *"I think 10 years from now, we'll probably go from seeing teams of 500 to teams of maybe 10-15."* (S3). This reduction is complemented by the compression of subsets within the field. *"I think we will be expected to do a lot more... that kind of argument to hire another designer, I think, is going to get weakened"* (S1). J3 has seen the impact and the significant reduction in team size at his company recently:

*"I feel that what will probably happen is that the AI will be able to create all these user flows and the aesthetic... But I imagine that the team would be cut significantly. I saw a lot of layoffs happen within my company when they moved over to [confidential product name]. Originally, they were doing everything in React native. I think we lost two-thirds of the development team because they moved into [confidential product name]."* (J3)

The compression of roles is not just about shifting team sizes but also about adding responsibilities. Designers are increasingly asked to perform cross-disciplinary work, managing design, product, development, strategy, and possibly more responsibilities. This shift challenges designers' self-efficacy—their confidence in their ability to take on broader responsibilities and succeed in a changing landscape. To handle more responsibilities, many interviewees noted that AI's ability to handle tasks like prototyping and iteration has freed them to focus on stepping up and engaging more deeply with strategy. This ability to transition from tactical to strategic thinking is closely tied to self-efficacy, as designers must trust their capability to shift beyond traditional roles. There was a belief among some of the research participants that the future of UX design will not be about "pixel pushing" but rather about becoming a holistic designer. S4 highlighted this evolution: *"You start to get into, okay, I'm a graphic designer, but I do UX UI as well... I think what you're going to see is we need to grow into more holistic digital designers that can have the ability to do visual, interaction, and experience design... And then to layer all that on, we have to continue to grow our ability to strategically think about the problems that we*

*are being asked to solve."* (S4) This shift changes how UX teams are structured in companies with bigger team sizes and specialized roles to more hybrid, cross-functional roles as AI reshapes the UX landscape. Designers with a strong sense of self-efficacy may see this shift as an opportunity to expand their influence and skill sets.

In contrast, those with lower self-efficacy may struggle to adapt to these growing demands. S2 stated: *"In terms of strategy, that's where we will be most needed or that's where we have the most impact, as it is really understanding human behavior and business strategy."* (S2) It appears many research participants have the view that future UX designers will focus less on tactical execution of design tasks and more on navigating complex human behaviors and aligning them with business strategies. *"The future of UX will shift from manual design skills to higher-level strategic thinking focused on user needs"* (S1). S3 also shared a hopeful perspective on design's strategic role:

*"Thank God for those designers who came before me that really pushed design to have a "seat at the table"... Hopefully, AI and the integration of it doesn't go purely generative and wipe us all out, which I don't think will happen, but in that case, I see us [focusing] more on strategy."* (S3)

This quote highlighted the view that AI can streamline production tasks while human designers remain essential for strategic and empathetic roles. This transition reflects a growing belief that design is increasingly integral to overall business strategy and the evolving landscape of design work, where human involvement may shift more toward strategy and facilitation, with AI handling the execution. As the industry evolves, many believe that collaboration between AI and human designers will foster innovation and creativity. This perspective reflects high self-efficacy, suggesting confidence in designers' ability to adapt to changing industry demands, embrace AI as a collaborative tool, and continue to play a strategic and empathetic role in

shaping user-centered designs. By leveraging the strengths, it is hoped that businesses can develop more impactful and user-centered designs that resonate with their audiences better.

### **Design for the Next Generation**

The research participants expressed the hope that AI advancements may become a natural extension of the next generation's lives and will reshape design paradigms. Today, designers are tasked with integrating AI into products and services and anticipating a future where AI fluency is the norm rather than the exception. S4 highlighted this shift, stating, *"You have to adapt with the changes, you have to adapt with this new generation. So, we have to leverage AI to improve. But this leveraging of AI is also making our skills and experience better."* (S4) This statement underscored a broader view that AI serves not just as a tool but as a catalyst for continuous learning and professional development.

Several research respondents described integrating AI into everyday life for the next generation as an expectation that everyone will eventually adapt. S4 continued to point out: *"That's ultimately who I'm going to be designing for by the time I get settled somewhere and grow in my career - people that are my age from the next generation. I think they'll have that AI background and exposure - that's something nobody's had before. It won't be a foreign concept to them, so it's not something they're going to have to learn to accept. It's just going to be something that is."* (S4)

This perspective suggests that future users will not need to be guided into accepting AI; instead, they will expect it as an inherent part of their user experience. Therefore, there is a belief

among designers that they need to begin facilitating the AI creation process. *"I think designers will, in some ways, maybe not be their entire job, but I think a portion of their job will be facilitating the AI creation process."* (S7). For designers, this evolution presents an opportunity to adapt their approach to align with changing user expectations and AI advancements. As AI becomes more integrated into everyday experiences, designers may need to consider how the next generation's familiarity with AI influences design priorities. The challenge is in balancing intuitive user experiences with emerging technological capabilities. This shift suggests a growing area between AI-driven insights and human creativity, with the potential to shape products that remain relevant and engaging for future users. This view demonstrates high self-efficacy, indicating that designers believe they can adapt to AI-driven changes, embrace new responsibilities, and actively shape the integration of AI into user experiences. Rather than feeling threatened by AI, designers with high self-efficacy believe they can facilitate its development, leverage its capabilities, and ensure that future products align with the expectations of AI-native users.

### **Leverage Human-Centricity**

As AI continues to change the UX design landscape, a key theme emerged: participants in this work expressed their views on the importance of adaptation, curiosity, and leveraging AI for professional growth – all factors closely linked to self-efficacy theory. Designers are encouraged to be open to learning, willing to experiment with new tools, and actively engaged with the technologies shaping their field. According to self-efficacy theory, individuals who believe in their ability to succeed in new or uncertain environments are more likely to engage in

proactive learning behaviors and persist in the face of challenges. S2 observed that "*[UX designers] are curious... we're going to question what AI puts out.*" (S2) This sense of curiosity and willingness to question AI outputs could be critical in maintaining the integrity of design work and ensuring that AI tools are used effectively to enhance, rather than replace, the human aspect of design.

While AI can generate design iterations and automate tasks, respondents felt it still lacks the emotional intelligence (EQ) central to creating user-friendly, intuitive designs. S4 commented, "*AI has high IQ but low EQ... so I think that will be a big area for us as UX designers to focus on the human interaction side of it.*" (S4). From a self-efficacy perspective, designers who perceive themselves as capable of bridging this gap between AI's technical capabilities and human-centered design may be more confident in their professional future. This perspective suggests that some designers see their role in continuing to be rooted in empathy, intuition, and understanding the intricacies of human behavior—areas that AI does not fully address.

Some participants described AI's ability to streamline repetitive tasks as an enabler for focusing on higher-level strategic work, potentially fostering professional growth and increasing competitiveness. For instance, S4 shared how AI tools helped them transition from a freelancer to establishing their agency, saying, "*What [AI has] helped me do, to be honest, is like, me personally, who am I? I'm nobody in the grand scheme of things. But that's allowed me to honestly go from being a freelancer to saying I have an agency.*" (S4) This view reflects how AI tools may enhance efficiency and allow designers to grow their businesses and expand their professional capabilities.

Similarly, S2 described how AI helped them appear more competitive globally: *"Because it's not just people in the United States. There are people in London, all over Europe, that are trying to adapt the same work mentality... it allows me to appear better and more competitive and I couldn't do that before."* (S2) This suggests that for some designers, AI may help improve productivity so designers can position themselves more effectively in a global, competitive landscape.

Participants generally described AI as a tool that complements rather than replaces their work. While some viewed AI as enhancing efficiency and creating new growth opportunities, others emphasized the continued importance of human creativity, empathy, and strategic thinking, which are central to UX designers' roles. From a self-efficacy standpoint, designers who feel capable of integrating AI into their workflows while maintaining their unique human skills are more likely to embrace these technological changes rather than resist them. As designers navigate these changes, they describe a need to adapt, continuously learn, and integrate AI into their processes while staying firmly rooted in human-centric design principles. This process of adaptation and mastery reinforces self-efficacy by allowing designers to gain confidence through experience, further shaping their ability to thrive in an AI-driven future. The hope is that the future of UX will be a future that blends human expertise with technological advancements, ensuring that designers remain at the forefront of innovation.

### **Deeper Dive into the Four Sources of Self-Efficacy**

Upon analyzing the data, patterns emerged connecting designers' perceptions of AI and the principles of self-efficacy theory. First, performance accomplishments can enhance

designers' confidence as they recognize the competence gained through successful interactions with AI tools. Second, vicarious experiences are illustrated by AI's mentoring role for junior designers, facilitating skill development and creativity. Moreover, AI's capacity can provide real-time feedback and encouragement traditionally fulfilled by peers and supervisors, exemplifying verbal persuasion. Finally, AI's impact on individuals' reflects a spectrum of emotions; while some expressed fear in navigating new technologies, others felt optimistic as they adapt to these innovative tools.

AI and self-efficacy can influence how designers approach and adapt to emerging technology. A designer's confidence in their ability to learn and apply AI tools may shape their willingness to integrate AI into their workflow and embrace changes in their role. Those with higher self-efficacy may view AI as an opportunity to enhance their creative process, while designers with lower self-efficacy may experience hesitation or uncertainty about its impact. Understanding these dynamics can provide insight into how designers navigate the evolving landscape.

### **Performance Accomplishment: Embracing the Role of AI**

One of the strongest themes that emerged from the interviews can be interpreted as seasoned UX designers can draw upon their extensive industry experience to navigate the evolving landscape of technology and design. A few experienced designers reflected on how the advent of AI technology mirrored past industry shifts, such as the rise of the internet and the digital age. This historical perspective gave them confidence and groundedness, knowing that adaptation and evolution are inherent to the UX design profession.

*“We kind of saw when the Internet, like when the web first started, people had the same fear because of that uncertainty as well. So I feel like we're kind of going through what we went through in the early 90s where there's so much uncertainty...It is almost like a maturation of society.” (S1)*

The sense of accomplishment from this participant's past experiences also showed resilience among senior designers. S1 witnessed the evolution of design practices and technologies from static websites to mobile-first design and now AI-driven tools. Their history of adapting to new technologies and methodologies gave them the confidence that they will also adapt to AI.

Participants also noted that ambiguity and change have always been a part of the UX field. They emphasized that the role of a UX designer often starts without clear answers. J2 shared that *“90% of [UX is] not knowing what's going on and figuring out a problem...that's just part of the iteration process.”* (J2) As projects typically begin with unknowns, and only through research and iterative design does the final solution take shape. This comfort with uncertainty can reinforce the designers' self-efficacy, as they approach AI not as a threat but as the next phase of the project's natural progression.

In contrast, junior designers expressed a higher degree of concern about job stability and the potential displacement by AI tools. Unlike their senior counterparts, they lacked the historical experience of surviving past industry shifts, which affects their confidence. Their unease aligned with mastery experiences—or the lack thereof—strongly influenced their self-efficacy. Junior designers were noted to view AI's rapid advancement with trepidation without a backlog of accomplishments to draw upon.

*“The future of AI does make me more nervous because, I mean, we have no idea what will be capable in a couple more years and how my role might be replaced entirely by AI.” (J3)*



However, designers hope that AI tools will help bridge this gap. For junior designers, AI can provide guidance and support when real-world experience is limited. Using AI as a mentor or partner, junior designers can gain small wins and build their own mastery experiences. These small accomplishments contribute to growing confidence, demonstrating that while AI might create uncertainty, it also presents an avenue for skill development and experiential learning.

### **Vicarious Experience: AI as a Mentor and Partner**

Vicarious experience involves learning by observing others' successes and applying those insights to build confidence and competence. For UX designers, this experience traditionally comes from collaborating with peers and getting feedback during design critiques. However, with the integration of AI tools into their workflows, designers can leverage AI as a virtual mentor and collaborative partner, mainly when human colleagues are not readily available.

Some designers described AI tools like ChatGPT as resources for generating ideas and substitutes for the collaborative experience gained through teamwork. These tools offer a way to simulate the back-and-forth exchange of ideas that would usually occur with colleagues. This capability was particularly beneficial for junior designers in this study as AI acted as a sounding board for validation and ideation.

*"AI is kind of that sounding board to help guide me... as a guiding stone to help give feedback in real-time." (S7)*

Senior designers from this study view AI as a productivity tool as it allows them to balance the output with their experience to execute tasks with incredible speed and accuracy. It serves as a "partner at work" (S5) for ideation, helping them maintain momentum even when their colleagues are unavailable. AI tools can enable designers to explore ideas and gather

diverse perspectives, allowing them to “ping pong ideas” and test their approaches against established solutions. This process reinforces designers’ confidence, reminding them that similar challenges have been overcome before.

Some junior designers find that AI can offer a critical starting point, providing a structured way to approach design challenges and identify blind spots. These designers sometimes lack the historical knowledge and experience senior designers rely on, making AI a valuable tool for gaining insights and learning from previous approaches. S6 highlighted that “*AI gives you a start, like how to think. It can show you some blind spots of things that I probably wouldn't think of.*” (S6). Using AI as a mentor can give less experienced designers who may not have direct access to senior colleagues a sense of direction. It allows them to build their self-efficacy by achieving small wins and reinforcing their sense of accomplishment. It also helps prepare them for collaborative work, allowing them to test ideas and refine their thinking before presenting to peers. As J9 noted, having access to AI enables them to have “*more productive conversations with other designers*” (J9) so that they can develop their skills and improve team communication. By simulating the guidance of a more experienced designer, junior designers feel more prepared and confident when facing real-world design problems.

For senior and junior designers, AI tools also facilitate vicarious learning by allowing them to tap into broader experiences. S7 explained:

*“Someone else has had this problem before, and they figured out a design solution. But sometimes when you're a younger designer, or you are in a team of one, it can be really hard to track down what that is.”* (S7)

Designers expressed that AI expands the designer's toolkit by providing quick access to prior solutions with diverse approaches. This capability can help generate new ideas and validate design decisions when traditional support systems are unavailable. By facilitating access to previously designed solutions, AI can help designers visualize successful outcomes, fostering confidence in their ability to replicate similar results. This process not only streamlines the design workflow but also encourages high self-efficacy through the informed application of past successes.

Designers in this study viewed AI as an "on-demand mentor," allowing them to engage in vicarious learning that might otherwise be inaccessible. This experience fosters a stronger sense of self-efficacy as designers internalize lessons from simulated collaborations and apply them to their unique design contexts.

### **Verbal Persuasion: AI as a Source of Encouragement**

Verbal persuasion is a critical element of self-efficacy, involving direct encouragement, constructive feedback, and affirmation from external sources. While traditional verbal persuasion in UX design often comes from managers, peers, and stakeholders during design critiques and feedback sessions, AI tools may be utilized in ways to offer real-time guidance and support.

Tools such as ChatGPT and Midjourney deliver prompt, unbiased responses to designers' inquiries, introducing a novel approach to verbal persuasion. Unlike vicarious experience, where designers gain insights by observing others, interaction with AI can foster direct engagement that bolsters confidence. Many participants remarked that the AI's responsiveness enhanced their assurance in their design decisions. For instance, S5 stated that AI serves to "*double-check*

*intuitions or hypotheses that I have"*(S5), providing validation and suggestions when necessary. The participant contributes to heightened confidence because AI diminishes the tendency to second-guess oneself.

For junior designers in this study, AI's verbal persuasion is valuable as they seek validation for their work but may not have direct access to seasoned colleagues. AI tools fill this gap, providing feedback that reinforces their sense of competence. S5 shared that AI *"has made me more efficient and at times more confident of my work."* (S5) Participants expressed a boost of confidence when approaching unfamiliar tasks, helping them trust their instincts and make decisions more assuredly. When designers use AI to review their work or generate new ideas, they receive a form of verbal persuasion free from bias or hierarchy, contributing to a more level playing field regarding feedback. J2 shared that AI's suggestions often validated their initial ideas and provided feedback to *"enable us to make more rounds of changes and be able to make even greater designs and even greater experiences"* (J2), reinforcing their sense of direction and encouraging them to explore further without hesitation.

For both junior and senior designers in this study, AI acts as a valuable source of verbal persuasion, enhancing self-efficacy by providing designers with a trusted "voice" that validates their decisions and supports their creative journey. Just like the findings from Haro Soler's (2021) study, AI can be the "caring teacher" who effectively builds the designer's self-efficacy through verbal persuasion. This dimension of AI empowers designers, allowing them to tackle challenges with greater confidence and fostering a positive mindset essential for navigating the ambiguity and complexity inherent in UX design projects.

## Emotional State: Anxiety or Optimism in the AI Age?

For UX designers, the rise of AI has elicited a broad spectrum of emotional responses, with junior and senior designers in this sample demonstrating contrasting feelings ranging from anxiety and stress to excitement and optimism.

For some junior designers in this study, AI's rapid integration into design workflows triggers anxiety and stress, primarily concerning job security. The fear that AI might replace their roles increased nervousness and anxiety. J5 shared: *"I'm a bit nervous about AI in terms of my role because I know that some jobs can be easily replaced by the advancements of AI."* (J5). Many of these designers are in the early stages of their careers, still developing foundational skills and professional confidence. J1 explained that *"I needed to up my game with ChatGPT because I realized that actually knowing how to ask questions helps me solve problems that actually will make you stand out a lot more."* (J1) The pressure to learn and adapt quickly can lead to stress. Such emotional responses can negatively impact self-efficacy, as prolonged anxiety may result in avoidance behaviors and a reluctance to engage with AI tools fully. A lack of historical perspective could also compound this stress. Junior designers who have not yet experienced the ebb and flow of technological advancements in the industry may fear AI as a disruptive force. Unlike the senior designers in this study, the junior designers seemed to struggle to contextualize AI as just another phase in the natural evolution of UX design. Without this context, their anxiety appeared amplified, leading to a more cautious or fearful approach to AI adoption.

Conversely, most senior designers in this study perceived AI with optimism and curiosity. Many described feeling excited about AI as a catalyst for efficiency and innovation.

Having witnessed previous technology shifts myself—such as the introduction of websites, adoption of digital design tools, the mobile-first design wave, and the evolution of cross-device experiences—senior designers in this study more often embraced technological advancements and expressed being ready to adapt to new tools as it is a natural part of the UX profession.

In my twenty years of experience as a UX designer, I have used different design tools every three to four years including Adobe Photoshop, Dreamweaver, Invision, Balsamiq, Sketch, XD, Figma, and now Canva. Designers tend to be more adaptive and can learn tools quickly as long as they have strong foundational design skills. The general view of the senior designers in this study is that AI is no different from any other tool that has emerged over the past two decades.

Another source of optimism among senior designers in this study is the evolving nature of their roles. Many are excited about the chance to "*have a seat at the table*" (S3) and contribute to strategic decision-making in the organization. As AI automates routine design tasks, senior designers appear to believe they will have more bandwidth to engage in higher-level thinking, influence business strategy, and shape the direction of products and services. S2 stated: "*Every business wants their business to grow. So any position related to [business growth] or the optimization [of business growth] are always going to have a future.*" (S2) This shift aligns with their aspirations to move beyond execution and become integral to guiding organizational strategy.

The contrast in responses between junior and senior designers suggests there may be a need for tailored support systems. Perhaps junior designers could benefit from mentorship and structured learning opportunities that demystify AI and frame it as a tool for collaboration and

growth. Providing them with guided experiences and small, manageable wins with AI could transform their stress into a more constructive, growth-oriented mindset. Senior designers have expressed that reinforcing their role as strategic leaders and providing training on the advancement of AI may help reduce their anxiety.

*"There's [no one] to teach you how to use it effectively. You kind of have to go in there and figure it out yourself... You can probably find a guide online...but [that is not] the right way to do it." (S1)*

This support can help designers feel more confident in their work amidst AI, showcasing how AI can empower them to make more impactful contributions and enhance their optimism. This would help maintain their positive emotional state, promoting a resilient and adaptive approach to change.

Understanding these emotional responses is crucial for fostering self-efficacy across all experience levels. While anxiety and stress are natural reactions to uncertainty, promoting a culture of learning and resilience can help junior and senior designers view AI not as a disruptive force but as a valuable tool in their ever-evolving toolkit.

## **Practical Implications and Discussion**

As AI becomes more integrated into UX design, the roles and expectations of designers are undergoing significant transformation. Designers are increasingly expected to balance creative problem-solving with AI-assisted efficiencies, leading to role compression where they must accomplish more with fewer resources. While automation can streamline repetitive tasks such as wireframing or prototyping, organizations must ensure that designers are not merely

burdened with additional responsibilities but are supported in adapting to this evolving landscape. Companies should reconsider team structures and workflows, leveraging AI to enhance—not replace—design’s strategic and human-centered aspects.

A key factor in how designers navigate this shift is self-efficacy—the belief in their ability to adapt and succeed. Senior designers in this study demonstrated a pragmatic outlook toward change, shaped by years of experience adapting to new tools and methodologies. This "que sera" attitude allows them to maintain confidence and resilience despite uncertainty. Rather than perceiving AI as a threat, those with higher self-efficacy are likely to recognize it as an opportunity to enhance productivity, streamline repetitive tasks, and focus on more strategic, creative aspects of their roles. This highlights an important implication - the more exposure designers have to AI in a constructive and empowering manner, the more they develop a sense of control over how they integrate it into their work. However, self-efficacy is not developed in a vacuum; organizations should actively foster it through structured programs that leverage Bandura’s four sources of self-efficacy: performance accomplishment, vicarious experience, verbal persuasion, and physiological state, which includes emotional state.

Performance accomplishment is the most effective way to build self-efficacy, as firsthand success reinforces an individual’s belief in their ability. To support adoption, organizations should provide hands-on AI training that allows designers to experiment with tools in a controlled, low-risk setting. Designers should be introduced to AI incrementally through workflow, starting with small, low-risk projects that allow them to gain confidence in using AI for augmentation. Creating structured upskilling programs where designers can receive dedicated time for career development ensures that AI competency grows at a manageable pace.



Additionally, organizing events like designathons can enable designers to solve challenges using AI to gain practical experience while fostering innovation in a safe environment.

Vicarious experience also plays an important role in increasing self-efficacy because it allows designers to see how AI adoption could lead to success. Organizations should create mentorship programs where experienced designers can share their industry knowledge and experience with technology disruption with junior designers to help them build confidence in AI. To further reinforce vicarious learning, companies should designate AI champions within each department—trusted team members who are responsible for identifying relevant AI tools, developing adoption roadmaps, and providing ongoing support to their colleagues. These structured peer learning mechanisms reduce uncertainty and make AI adoption feel more accessible.

Verbal persuasion helps reinforce designers' belief that they can successfully use AI tools, particularly when paired with performance accomplishments and vicarious learning. Organizations must provide regular feedback and positive reinforcement, recognizing and celebrating employees who effectively integrate AI into their workflows. This can be reinforced through formal recognition programs, awards, or incentives for AI adoption. Additionally, hosting peer learning workshops where designers can share successes, challenges, and insights promotes a culture of AI exploration while normalizing its use. AI proficiency should also be formally embedded into career development structures; organizations can update their Career Ladder Matrix (CLM) to include AI-related skills as core competencies for advancement, signaling that AI fluency is a key factor in professional growth.

Finally, emotional state influences how designers emotionally respond to AI adoption. Anxiety and uncertainty can inhibit engagement, making it essential for organizations to create an environment where designers feel psychologically safe. Clearly communicating the company's AI policy—including its role as an augmentation tool rather than a replacement—helps alleviate concerns about job displacement. Additionally, ensuring a human-centered approach to AI integration reassures designers that AI is designed to enhance their creativity and strategic thinking, not diminish their contributions. Addressing AI-related fears through transparent discussions about job security and the evolving role of UX designers can further mitigate resistance. Above all, organizations must cultivate a culture of experimentation where designers feel comfortable exploring AI without fear of negative consequences, allowing them to develop confidence in their ability to integrate AI into their work.

By proactively incorporating these four dimensions into AI training and organizational support structures, companies can ensure that AI adoption is not just a technological shift but a transformation that empowers designers to thrive. Performance accomplishment builds mastery through hands-on experience; vicarious learning helps designers learn from their peers; verbal persuasion reinforces AI proficiency as a valued skill; and a supportive environment that considers their emotions will increase the chance that designers feel secure in their evolving roles. Organizations that take a structured, evidence-based approach to AI training will cultivate a workforce that is confident, adaptable, and capable of leveraging AI as a tool for innovation rather than a source of disruption.

## **Limitations and Future Research**

This study has several limitations that should be considered when interpreting the findings. First, while providing rich qualitative insights, the sample size of 20 participants limits the generalizability of the results. This study is intended as an exploratory lens to guide future research rather than a basis for making broad predictions or generalizations about the evolving role of UX designers.

Second, all interview participants were based in the United States and Canada, limiting the study's geographic and cultural diversity. The design and technology industries vary widely across global markets, and regional differences in AI adoption, workplace culture, and industry practices may influence how designers perceive the impact of AI on their roles. Further studies involving participants from different regions and industries could reveal whether these findings hold in diverse contexts.

Also, the rapidly evolving nature of AI technology presents a significant limitation. As new tools and capabilities emerge, designers' perceptions of AI and its influence on their roles will likely shift. This study offers a snapshot of the current attitudes and experiences, but the findings may become outdated as the technology and its applications advance. Future research could benefit from a longitudinal approach, tracking changes in self-efficacy and role adaptation over time to provide a more dynamic understanding of AI's long-term impact on the design field.

In addition, this research focused exclusively on UX designers, excluding other key roles within cross-functional teams, such as product managers and developers. As a result, the study may primarily reflect designers' aspirations and expectations rather than a holistic view of how design roles might evolve in practice when collaborating with diverse disciplines. Future

research could expand the scope to include a broader range of roles, offering a more comprehensive perspective on how AI might reshape team dynamics and responsibilities. It also did not explore how organizational factors—such as company culture, leadership strategies, and internal policies—might influence designers' adoption of AI tools. These factors could significantly affect the extent to which AI becomes integrated into design workflows and how roles evolve in practice. Future research could investigate how organizational support and structure contribute to designers' self-efficacy and role evolution.

While this study highlights how UX designers currently perceive AI, future research should explore how design education can better prepare students for future AI-integrated workplaces. Many junior designers in this study expressed uncertainty and anxiety about AI's role in their careers, often feeling ill-equipped to leverage these tools effectively. Given the rapid advancements in AI, design curricula must evolve to ensure that emerging designers develop the technical proficiency and strategic thinking necessary to work alongside AI. Future studies could investigate how universities can better incorporate AI literacy into their programs and whether current educational models adequately address the skills designers need. Research could also explore best practices for teaching AI integration in UX, such as hands-on projects with AI-driven design tools and case studies that showcase successful AI adoption in the industry.

In short, while this study provides valuable insights into how AI affects UX designers' self-efficacy and role perceptions, the limitations highlight the need for further research to validate and expand upon these findings in a broader and more dynamic context.

## Conclusion

Adopting new technology is rarely a straightforward process—it is an evolving journey shaped by curiosity, hesitation, and, in most cases, eventual acceptance. This journey is not the same for each person or each technology. In this study, I examined how UX designers feel as AI becomes more prominent in their roles, uncovering the uncertainties and opportunities that come with this shift. By exploring their perspectives, I attempted to understand how AI is reshaping their responsibilities, influencing their career trajectories, and potentially affecting their feelings and sense of self-efficacy. This research highlighted factors that are influencing designers' ability to integrate AI into their work, illuminating how experience, training, and mindset can play crucial roles in shaping their experience and emotional responses to technology adoption.

While this study captured designers' reflections at a specific moment in time, their feelings reflected familiar patterns of navigating change. My own experience with GPS reminded me of the journey many designers are currently navigating with AI. I vividly recalled the thrill of getting my driver's permit in the 90s, relying on crinkled paper maps to chart my course. When GPS first emerged, I hesitated to trust a disembodied voice directing my every turn. My skepticism deepened when a GPS misdirection nearly led me straight into the Hudson River. Over time, as the technology matured and I became more familiar with it, GPS shifted from being a novelty to an indispensable part of my daily life. Though AI is far more complex and wide-reaching than GPS ever was, some of the thoughts and feelings behind the adoption—the hesitation, experimentation, and gradual integration—mirrored some of the feelings designers expressed in this study as they encountered AI in their work.

This study indicated that the UX designers involved experienced a range of reactions from initial skepticism to experimentation and, in some instances, a growing reliance on AI. Since many of the junior designers in the study are still in the process of developing their skills, it is not surprising that they expressed feelings of uncertainty and voiced concerns that AI could potentially undermine their roles before they had the opportunity to fully define them. Conversely, several senior designers in this study, drawing on their years of experience with new tools, conveyed a greater sense of confidence. They viewed AI as a means to offload repetitive tasks, thereby creating more room for strategic and creative contributions.

Through the lens of Bandura's self-efficacy theory, the comments from the designers I interviewed highlighted how their confidence in their ability to adapt shaped their perspectives on AI. Whether through past experiences of mastering new tools (performance accomplishment), learning from observing peers (vicarious learning), encouragement from others (verbal persuasion), or managing their own emotional responses, many described strategies that help them build resilience as they face the ongoing changes AI introduces into their field.

This research suggests the importance of supporting designers' sense of self-efficacy by integrating Bandura's four sources into AI training. This may help ease future transitions and foster more positive AI adoption experiences. This study offers an initial step in the ongoing effort to understand how designers navigate technological change and how they can continue to adapt, iterate, and innovate—regardless of the challenges that lie ahead.

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## Appendix

### APPENDIX A: INTERVIEW QUESTIONS

| #   | Question                                                                                                                                                                                                                    | Objective                             |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| 1.  | Can you tell me about your roles and responsibilities in your current or most recent position?                                                                                                                              |                                       |
| 2.  | Have you used AI in your most recent role? What are your experiences?                                                                                                                                                       | RQ1: Thoughts and feelings towards AI |
| 3.  | What are your thoughts and feelings about AI? It is fine to share a mix of positive and negative emotions, it doesn't have to be all one or all the other.                                                                  | RQ1: Thoughts and feelings towards AI |
| 4.  | What are your thoughts and feelings about the <b>future</b> of AI? Again, it is fine to share a mix of positive and negative emotions, it doesn't have to be all one or all the other.                                      | RQ1: Thoughts and feelings towards AI |
| 5.  | What are the benefits and drawbacks of AI, if any?                                                                                                                                                                          | RQ4: Risks and benefits               |
| 6.  | How has AI changed your work as a designer thus far?                                                                                                                                                                        | RQ2: Role of a designer – present     |
| 7.  | How do you think AI will change your work as a designer in the future?                                                                                                                                                      | RQ2: Role of a designer – future      |
| 9.  | In terms of the future of AI in UX design, what do you think are the potential risks and benefits?                                                                                                                          | RQ4: Risks and benefits               |
| 10. | Anything else you want to share?                                                                                                                                                                                            |                                       |
|     | As AI continues to advance in the field of UX design, do you feel that your success will increasingly rely on your ability to leverage AI effectively or your own skilled experience remain the primary drivers of success? |                                       |